Interspeech 2024
1-5 September 2024, Kos, Greece

A comparative analysis of sequential models that integrate syllable dependency for automatic syllable stress detection

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Abstract

Automatic syllable stress detection is typically operated at syllable level with stress-related acoustic features. The stress placed on a syllable is influenced not only by its own characteristics but also by its context in the word. However, traditional methods for stress detection overlook the contextual acoustic factors that influence stress placement. By addressing this issue, we study sequential modeling approaches by integrating the syllable dependency for automatic syllable stress detection using a masking strategy. This approach considers a sequence of syllables at the word level and identifies its stress label sequence. We explore various sequential models, such as RNNs, LSTMs, GRUs, and Attention networks. We conduct experiments on the ISLE corpus comprising non-native speakers speaking English. From the experiments, we observe a significant improvement in the performance with all sequential models compared to the state-of-the-art non-sequential baseline (DNN).

Index Terms: syllable stress detection, sequential models

1. Introduction

In stress-timed languages like English, every word has one syllable that is more prominent or relatively emphasized, than any other syllable in the same word and is termed a stressed syllable [1]. To make the stressed syllable stand out from its neighbors, it is produced with greater physiological effort during the speech production process. Stressed syllable carries the meaning of the word, and misplacement can lead to confusion in understanding. Due to native language influence, second language (L2) learners of English often face difficulty in understanding and placing stress on correct syllables and make errors. To provide personalized guidance to L2 learners, computer-assisted language learning (CALL) [2] systems have been developed with an automatic syllable stress detection module being a key component.

Automatic syllable stress detection involves classifying syllables into two categories (stressed/unstressed), making it a two-class classification problem. Many studies in the literature tried analyzing the acoustic parameters responsible for stress [1]. From their analysis, it is claimed that intensity, duration, and pitch are the three main acoustic cues that vary among stressed and unstressed syllables [3]. Several heuristic-based features are computed over these acoustic cues and used for automatic syllable stress detection task [4, 5, 6]. In [5], they have incorporated the context while computing the acoustic features. In [6], explicit context features were proposed highlighting the importance of considering context in syllable stress detection. Many machine learning (ML) models like boosting, bagging, decision trees, and support vector machines were used over the heuristic-based features for stress detection task [7]. In native speakers, stress cues such as higher intensity, duration, and pitch make stress and unstressed syllables easily distinguishable using heuristic-based features. However, these cues may vary in L2 learners due to native language influence, leading to errors in heuristic features which in turn, pose challenges to the standard machine learning algorithms in stress detection tasks [8]. To address this issue, many deep learning models were used highlighting the need for representation learning, and have shown significant improvement over traditional ML models. Tian et al. [9] used an attention-based neural network and bidirectional LSTMs for stress detection problem using Mel frequency cepstral coefficients (MFCCs), energy, and pitch features. Ruan et al. [10] performed stress detection using a transformer network. In [8], a novel methodology was introduced using variational autoencoders for stress detection. In [11], it is stated that the phenomena of stress is suprasegmental, which means it can be affected by the features within the syllable or multiple syllables in a word for creating the perception of only one syllable being emphasized. The duration or elongation of vowel sounds in a syllable plays a crucial role in creating the perception of stress. Interestingly, in few scenarios, even though the vowel sound is short, emphasis may still be present perceptually but is not captured in the acoustics which could be because of the context and highlights the influence of the context in stress perception. According to the theory of articulatory phonology [12, 13], the stressed syllables tend to maintain their identity (resist coarticulation with an adjacent segment in an unstressed syllable) whilst unstressed segments across the syllable boundary are disproportionally affected by articulation. This suggests that incorporating the reflection of articulation effects in acoustics of the contextual syllables is helpful in enhancing stress detection models. Despite these insights, the literature has predominantly concentrated on linguistic-based context features and lacks studies that explore contextual acoustic features.

In all the above mentioned studies, the problem of stress detection has been modeled as sequence-independent focusing at syllable level. However, literature suggests that context plays a crucial role in stress perception. Building upon this observation, we hypothesize: modeling stress detection as a sequence dependent problem could potentially enhance stress detection performance. Unlike existing approaches, modeling stress detection at word level, considering the sequence of syllables within each word helps in learning the pattern of syllable stress across syllables in a word. Also, it allows capturing the interdependencies among syllables and leverages contextual information.

In this work, we study sequence-dependent stress detection modeling using four different deep learning based sequence modeling architectures such as GRUs, RNNs, LSTMs, and Attention networks. To accommodate sequences of varying lengths, we employ a masking strategy, ensuring robustness across different input sizes. We compare the proposed ap-

10.21437/Interspeech.2024-2404

proach with a sequence-independent modeling state-of-the-art approach i.e., simple deep neural network (DNN) [8]. We perform experiments on the ISLE corpus which consists of polysyllabic English words uttered by non-native speakers of German, and Italian. We explore three distinct sets of features for stress detection: 1) state-of-the-art heuristics-based acoustic features, 2) a combination of acoustic and context features, and 3) self-supervised representations (Wav2Vec-2.0). From the experimental results, we observe that there is a significant improvement of 5.47%, 6.21%, 6.63%, and 6.81% in classification accuracies with RNNs, LSTMs, GRUs, and Attention networks compared to the sequence-independent model (DNN).

2. Dataset

For experiments, we use the ISLE corpus [14], which contains speech utterances from 46 non-native English speakers out of which 23 are German (GER) and 23 are Italian (ITA). Each speaker contributes approximately 160 sentences. We employ automatic force alignment to obtain phoneme alignments for each utterance, followed by manual correction by a team of five linguists to ensure accuracy in reflecting the speakers' pronunciation nuances. Using the P2TK syllabification software [15], we derive syllable alignments from the phoneme alignments. The same linguist team manually annotated syllable stress labels, ensuring each word contains only one stressed syllable. For experiments, we focus on polysyllabic words with two or more syllables, as per the methodology outlined in [6], resulting in 12,388 stressed and 16,005 unstressed syllables. We create train and test splits for both the GER and ITA datasets, maintaining balance across speakers' nativity, age, sex, and proficiency levels, following the work in [14]. The distribution of stressed and unstressed syllables in both the GER and ITA datasets remains relatively balanced across the train and test splits.

3. Methodology

Figure 1 illustrates the entire process of the sequence-dependent modeling approach for studying stress detection task [1]. This involves four major stages: 1) Data processing, 2) Masking, 3) Sequence modelling, and 4) Post-processing. We describe the four stages in the following subsections.

3.1. Data processing:

Given a speech signal of a word (with 'n' syllables), we segment it into 'n' syllables based on the start and end timestamps obtained from the syllable time-alignments. Features of 'd' dimension are computed for each syllable separately. These features are concatenated as a sequence of syllable features resulting 'nxd' feature for the entire word on 'n' syllables. Due to the variability in the number of syllables across words, we employ upsampling by padding the syllable feature sequence. This involves padding the sequence with a 'd' dimensional vector consisting of a random integer (we consider it as '-1'), for 'r' number of times. This process transforms the feature representation from 'nxd' to '(n+r)xd'. The variable 'r' varies for each word to match the highest syllabic word in the dataset.

3.2. Masking:

These sequences of syllable features are fed to a masking layer before training the sequential model. This masking layer is configured to recognize the random integer used in upsampling and it effectively conceals the padded features within each sequence, thereby preventing their contribution in the model training.

[1] `https://github.com/jhansimallela/Interspeech2024_SequentialModeling`

3.3. Sequence modelling:

Following the masking layer, the word-level sequence of syllable features is passed through the sequential model for sequence classification. As shown in the block diagram, the sequence of syllable features is fed into each of the sequential models individually and trained. The loss is computed only for the unmasked syllable sequence and backpropagated to learn the dependencies among the sequence.

Sequential models are designed to capture temporal dependencies within sequences, making them well-suited for several speech applications. In this work, we consider four different sequential models to validate the proposed approach.

1) Recurrent Neural Networks (RNNs): A class of neural networks designed for sequential data processing [16], RNNs maintain hidden states that capture temporal dependencies between elements in the sequence. However, simple RNNs has the problem of vanishing gradients.

2) Gated Recurrent Units (GRUs): A variant of RNNs [17], GRUs use gating mechanisms to control the flow of information within the network, facilitating better long-term dependencies modeling while mitigating the vanishing gradient problem.

3) Long Short-Term Memory Networks (LSTMs): Another type of RNN, LSTMs [18] incorporate memory cells and various gates (input, output, forget) to selectively retain or discard information over time, enabling more effective handling of long-range dependencies and mitigating issues like vanishing gradients.

4) Attention Networks: Unlike traditional RNNs, attention mechanisms [19] allow models to focus on specific parts of the input sequence, dynamically weighting the importance to the elements of the input sequence for predicting each output. This enables more effective learning and reasoning, particularly in tasks where certain elements are more relevant than others for a given element in the output sequence.

Typically, each word in English consists of only one primary stressed syllable. However, this constraint is not considered in traditional syllable-level stress detection models. In this work, since the loss is computed across all the syllables in a word, it indirectly encourages the model to predict only one syllable as stressed and the rest as unstressed, by incorporating the constraint and addressing the issue.

3.4. Post-processing:

Following model training, we conduct a post-processing step (non-learnable) to ensure that each word contains only one stressed syllable. In this step, among the stressed syllable probabilities from the classification model, we consider the one with the highest probability as stressed and the others as unstressed.

4. Experimental Setup

In this section, we first brief about the features considered for the experiments – 1) heuristic-based acoustic and context features, 2) self-supervised wav2vec 2.0 representations. Next, we discuss about the architectural details of the sequential models used in the proposed approach.

4.1. Acoustic and Context features:

Acoustic features (A): Typically, the syllable stress depends on three prominence measures: intensity level, duration, and pitch – around the sound unit with the highest sonority in the respective syllable. Acoustic features are computed using cues from the feature contours (e.g. short time energy contour) representing these prominence measures. However, short-time energy contours have large variabilities across the unstressed syllables

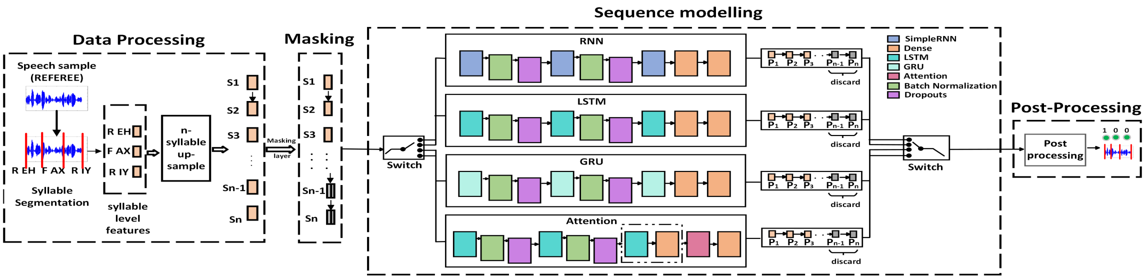

Figure 1: *Block diagram of the sequence-dependent modeling approach for syllable stress detection*

and often introduce peaks based on the surrounding sounds in the speech context [6]. By addressing this issue, a sonority-based feature contour is proposed by combining the sonority-motivated [20] cues with short-time energy contour. With the motivation of the work in [6] from the sonority-based feature contour, we also compute 19-dimensional statistical features and are termed as acoustic features(A) in further sections.

Context features (C): As mentioned earlier, context plays a crucial role in the perception of stress. In [21], they have proposed 19-dimensional binary context feature vector for each syllable. These context features reflect the information about the syllable nucleus type, category of phoneme preceding and following the syllable nucleus, and word position with respect to the pause in the sentence.

4.2. Self-supervised representations (Wav2Vec 2.0):

In recent times, self-supervised models for speech representation, notably Wav2Vec 2.0 [22] and Hubert [23], have gained considerable attention. These models offer a novel approach to learning meaningful representations from speech data without requiring manual labeling. Wav2Vec 2.0, in particular, has emerged as a widely adopted framework due to its versatility across various speech-related tasks [24, 25]. By directly processing raw speech data, it captures a wide range of information, including spectral features, phonetic details, speaker characteristics, and even semantic nuances. Considering this, in this work, we use 768-dimensional Wav2Vec 2.0 features for our experiments.

4.3. Architecture details:

The RNN model has three recurrent layers [cells: 64, 32, 16] followed by two dense layers [hidden units: 4, 1]. GRU consists of three gated recurrent units [cells: 64, 32, 16], followed by two dense layers [hidden units: 4, 1]. The LSTM model has three LSTM layers [cells: 64, 32, 16] and two dense layers [hidden units: 4, 1]. For the attention model, we consider three LSTM layers [cells: 64, 32, 16] followed by a dense layer [hidden units: 4]. The LSTM with 16 cells is used as the query, and the dense layer with 4 units is used as the value. They are passed to an attention layer with query and value as inputs, which is then connected to a dense layer of 1 unit for classification. The dense layers use *Relu* as an activation function and *Adam* as an optimizer. We perform the experiments in a five-fold cross-validation setup in a round-robin fashion.

Baseline: As a baseline, we consider the state-of-the-art syllable level stress detection model i.e simple DNN model with 5 layers [hidden units: 64, 32, 16, 4, 1]. Binary cross-entropy is the loss function for all models. In [8], the authors considered the representation learning-based models besides simple DNN. Though they showed that the representation learning based models outperformed simple DNN, in this work, simple DNN is chosen for fair comparison as proposed sequential models do not include any explicit representation learning framework mentioned in [8]. We believe that the similar conclusion can be drawn between the sequential and sequence-independent models incorporated with representation learning based framework. All the parameters mentioned above are obtained by maximizing the performance on the validation set.

We perform experiments in three different scenarios: **Matched:** Individual models are trained using the GER and ITA train sets, and then evaluated independently on the GER and ITA test sets. **Combined:** A single model is trained using combined data from both GER and ITA, and then evaluated separately on the GER and ITA test sets. **Cross:** Models are trained using the GER (ITA) train set and tested using the ITA (GER) test set.

From the literature, it is evident that incorporating context features are enhancing the performance of syllable stress detection task. In this study, we aim to validate this claim using sequential models and assess the extent of the impact of context features. To investigate this, we conduct experiments both with and without the addition of context features to the acoustic features. Hence, we experiment with three sets of features. 1) Using only acoustic features, 2) Using the combination of acoustic and context features, and 3) Using Wav2Vec 2.0 features

5. Results and Discussion

In this section, we analyze the results obtained from three sets of features: 1) Acoustic (A), 2) Acoustic plus context (A+C), and 3) Wav2Vec 2.0. We examine these results under three scenarios: matched, combined, and cross, for both the GER and ITA cases. We compare the performance of all sequential models (RNNs, GRUs, LSTMs, and Attention networks) with that of the non-sequential model baseline (DNN). We present a comparison of classification accuracies with and without postprocessing. Additionally, we evaluate the effectiveness of learned representations from sequential models and compare them with the DNN using t-SNE plots of the feature space in each model.

Table 1: *Accuracies of DNN, RNN, LSTM, GRU and Attention models with and without(in bracket) postprocessing considering Acoustic features under three different scenarios.*

A		DNN	RNN	LSTM	GRU	Attention
GER	**Matched**	84.35 (81.15)	89.13 (88.05)	89.99 (88.73)	89.72 (88.83)	**90.01 (88.97)**
	Combined	85.27 (82.58)	89.27 (88.597)	90.157 (89.15)	90.68 (89.44)	**90.72 (89.85)**
	Cross	83.31 (78.55)	87.37 (87.20)	88.41 (87.6)	88.38 (87.47)	**88.66 (87.93)**
ITA	**Matched**	84.67 (81.85)	90.08 (89.54)	90.58 (89.83)	90.70 (89.55)	**90.99 (90.05)**
	Combined	85.56 (82.61)	90.31 (89.62)	90.74 (89.81)	91.47 (90.39)	**91.84 (90.34)**
	Cross	83.78 (79.96)	89.04 (88.04)	89.99 (88.91)	90.41 (89.28)	**90.59 (89.83)**

5.1. Comparison across three different sets of features:

Using Acoustic features (A): Table 1 presents the classification accuracies with and without(in brackets) post-processing obtained from DNN, RNNs, GRUs, LSTMs, and Attention using Acoustic features under three scenarios with both GER and ITA. From the table, it is observed that sequential models consistently outperform the baseline, with the Attention network achieving the highest accuracies in all cases. In the matched scenario, the Attention model stood out with the highest accu-

Table 2: *Accuracies of DNN, RNN, LSTM, GRU and Attention models with and without(in bracket) postprocessing considering A+C features under three different scenarios.*

A+C		DNN	RNN	LSTM	GRU	Attention
GER	**Matched**	92.80 (88.61)	93.75 (92.98)	94.32 (93.40)	93.98 (93.06)	**94.86 (93.45)**
	Combined	92.85 (89.27)	94.62 (93.37)	94.94 (93.95)	94.79 (93.83)	**95.8 (95.01)**
	Cross	88.44 (85.52)	91.13 (89.79)	91.88 (90.23)	91.05 (89.58)	**91.95 (90.59)**
ITA	**Matched**	91.32 (88.29)	92.94 (92.20)	93.45 (92.55)	93.83 (92.91)	**94.00 (93.33)**
	Combined	92.70 (89.49)	94.06 (93.03)	95.69 (94.43)	95.38 (93.85)	**95.75 (95.67)**
	Cross	91.02 (86.49)	92.37 (91.1)	92.53 (91.48)	92.53 (91.58)	**92.79 (92.21)**

racies indicating an absolute improvement of 5.66% and 6.32% for GER and ITA, respectively, when compared to the DNN. Similarly. in the combined scenario, the Attention model outperformed all the models with accuracies of 90.72% for GER and 91.84% for ITA, and absolute improvements of 5.45% and 6.28% for GER and ITA, respectively when compared to the DNN model. Again in the Cross scenario, the Attention model showed an absolute improvement of 5.35% and 6.81% for GER and ITA, respectively, compared to the sequence-independent baseline model. Overall, these findings underscore the effectiveness of all sequential models, particularly the Attention network, in enhancing syllable stress detection accuracy.

Using Acoustic plus Context features (A+C): Table 2 presents the classification accuracies obtained for all sequential models and DNN using both acoustic and context features. From the results, it is observed that in A+C, there is a similar pattern as in acoustic features(A) among the improvements in the sequential models over DNN. All sequential models are outperforming the baseline with significant improvements. Compared to using only acoustic features, the combination of acoustic and context features has improved the classification accuracies significantly across all scenarios proving the context plays a crucial role in stress detection. Among all sequential models, the highest accuracies are with attention network, with absolute improvements of 2.06% and 2.68% & 2.95% and 3.05% & 3.51% and 1.77% for GER and ITA under matched & combined & cross scenarios, respectively compared with DNN.

Using Wav2Vec 2.0 representations: Table 3 presents the classification accuracies of all sequential models and DNN using Wav2Vec 2.0 feature representations. From the results, it is observed that among all sequential models, only GRU is performing better compared to the baseline under without postprocessing case. After postprocessing, the baseline is outperforming all sequential models. This suggests that the postprocessing step is showing a huge impact in the non-sequential model (DNN) but not in the sequential models when we consider Wav2Vec 2.0 representations. Unlike acoustic and context features, which inherently encode stress-related sequential patterns in the speech signal, Wav2Vec 2.0 features represent a high-dimensional and abstract representation of various speech characteristics. This could be the reason that sequential information is not well captured across the syllables in Wav2Vec 2.0 feature representations as in acoustic and context features. Additionally, the performance of the DNN may indicate that simpler, non-sequential models can better leverage the rich feature representations provided by Wav2Vec 2.0. These models might excel at extracting relevant patterns and information directly from the high-dimensional feature space without the need for explicit sequential modeling. However, it's important to note that despite the DNN's superior performance with Wav2Vec 2.0 features, the sequential models still outperform it when utilizing acoustic and context features, highlighting the importance of acoustic and context features.

Table 3: *Accuracies of DNN, RNN, LSTM, GRU and Attention models with and without(in bracket) postprocessing considering Wav2Vec-2.0 features under three different scenarios.*

Wav2Vec-2.0		DNN	RNN	LSTM	GRU	Attention
GER	**Matched**	**94.04** (91.32)	92.81 (91.09)	93.12 (91.70)	93.27 **(92.36)**	91.56 (90.74)
	Combined	**94.09** (92.93)	93.59 (92.17)	93.77 (92.36)	93.8 **(92.96)**	92.46 (91.07)
	Cross	**90.74** (87.33)	89.6 (88.18)	89.98 (88.31)	90.19 **(88.76)**	87.63 (86.72)
ITA	**Matched**	**93.61** (91.13)	92.07 (90.86)	93.14 (91.68)	93.46 **(91.79)**	92.29 (91.13)
	Combined	**94.33** (92.26)	93.69 (91.79)	93.79 (92.15)	93.85 **(92.68)**	93.11 (91.55)
	Cross	**92.09** (88.89)	90.73 (89.89)	91.369 (90.41)	91.79 **(90.86)**	89.94 (89.46)

Effect of postprocessing: We mentioned earlier that the proposed approach indirectly encourages the sequential model to predict only one syllable as stressed and others as unstressed which is not the case in the non-sequential models (baseline). Tables 1, 2, and 3 reveal that while the postprocessing step significantly impacts DNN performance, its effect is minimal on sequential models. This highlights the ability of sequential models to learn syllable dependencies effectively, limiting the need for postprocessing.

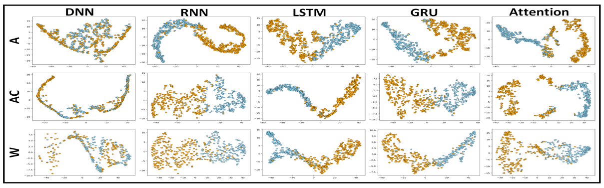

Figure 2: *t-SNE Visualizations of representations learned from DNN, RNN, LSTM, GRU and Attention models with Acoustic(A), Acoustic+Context(AC) and Wav2Vec-2.0(W) features (blue-stressed, orange-unstressed)*

5.2. Visualization of representations using t-SNE plots

Figure 2 depicts t-SNE visualizations of representations learned from DNN, RNN, LSTM, GRU, and Attention models across all three feature sets. These visualizations show clearer separation between stressed and unstressed syllable representations in sequential models compared to DNN. This stands true across all feature sets (A, A+C, and Wav2Vec 2.0). Despite DNN achieving higher results with Wav2Vec 2.0, the visual representations suggest that clear separation occurs in sequential models but not in DNN, indicating the importance of post-processing for higher accuracies in DNN under Wav2Vec 2.0. This supports our hypothesis emphasizing the necessity of a sequential modeling approach in the syllable stress detection task.

6. Conclusion

In this study, we explored a sequence-dependent modeling approach for syllable stress detection using advanced heuristic features (Acoustic and Context features) and Wav2Vec-2.0 feature representations. We evaluated four different sequential models (RNNs, GRU, LSTMs, and Attention network) and compared their performance against the state-of-the-art non-sequential approach (DNN). Our experiments on the ISLE corpus demonstrated that the sequence-dependent modeling approach consistently outperforms the baseline, highlighting the significance of acoustic and contextual syllable information for accurate stress detection. In the future, we would like to investigate unsupervised approaches for sequence-dependent syllable stress detection to overcome the difficulty in manual labeling of stress markings.

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